

SQL SET Operations

06 · SET Operations — Complete Reference Guide

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1. Rules of SET Operations

Rule 1 — Column Count: Both queries must have the same number of columns.

Rule 2 — Data Types: The data types of columns in each query should match.

Rule 3 — Column Order: The order of columns in each query must be the same.

Rule 4 — Column Aliases: Column names are determined by the FIRST SELECT statement.

Rule 5 — Correct Columns: Ensure correct columns are used to maintain data consistency.

→ *Example: Column count mismatch — this will FAIL*

```
SELECT FirstName, LastName, Country
FROM Sales.Customers
UNION
SELECT FirstName, LastName          -- Missing 3rd column
FROM Sales.Employees;
```

→ *Example: Column aliases taken from the FIRST SELECT*

```
SELECT CustomerID AS ID, LastName AS Last_Name
FROM Sales.Customers
UNION
SELECT EmployeeID, LastName
FROM Sales.Employees;
```

2. UNION — Combine Results, Remove Duplicates

→ *Task 1: Combine Employees and Customers into one table (no duplicates)*

```
SELECT FirstName, LastName
FROM Sales.Customers
UNION
SELECT FirstName, LastName
FROM Sales.Employees;
```

3. UNION ALL — Combine Results, Keep Duplicates

→ *Task 2: Combine Employees and Customers including duplicates*

```
SELECT FirstName, LastName
FROM Sales.Customers
UNION ALL
SELECT FirstName, LastName
FROM Sales.Employees;
```

4. EXCEPT — Rows in First Query NOT in Second

→ *Task 3: Find employees who are NOT customers*

```
SELECT FirstName, LastName
FROM Sales.Employees
EXCEPT
SELECT FirstName, LastName
FROM Sales.Customers;
```

5. INTERSECT — Rows That Exist in BOTH Queries

→ *Task 4: Find employees who are also customers*

```
SELECT FirstName, LastName
FROM Sales.Employees
INTERSECT
SELECT FirstName, LastName
FROM Sales.Customers;
```

→ *Task 5: Combine Orders and OrdersArchive without duplicates*

```
SELECT
    'Orders'          AS SourceTable,
    OrderID, ProductID, CustomerID, SalesPersonID,
    OrderDate, ShipDate, OrderStatus, ShipAddress,
    BillAddress, Quantity, Sales, CreationTime
FROM Sales.Orders
UNION
SELECT
```

```
'OrdersArchive' AS SourceTable,  
OrderID, ProductID, CustomerID, SalesPersonID,  
OrderDate, ShipDate, OrderStatus, ShipAddress,  
BillAddress, Quantity, Sales, CreationTime  
FROM Sales.OrdersArchive  
ORDER BY OrderID;
```